Exercise 2

01 THE STUDENTS TABLE

01 SELECT DISTINCT department
FROM students;

Output: department
IT
HR
Finance

02 SELECT department,
AVG(age) AS average_age
FROM students
GROUP BY department;

Output: department average_age
IT 20.5
HR 22
Finance 23

03 SELECT department, ~~students_id~~ COUNT(student_id) AS student_count
FROM students
~~WHERE~~ GROUP BY department
HAVING COUNT(student_id) >1;

Output: department student_count
~~04~~ IT 2
HR 2

04

```
SELECT student_id, name, age, department
  FROM students
  WHERE age BETWEEN 21 AND 23;
```

Output

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

Q5

```
SELECT student_id, name, age, department
  FROM students
  WHERE department IN (IT, HR)
   AND age > 21;
```

Output:

student_id	name	age	department
2	Bob	22	HR
~~4~~	~~Diana~~	~~23~~	~~Finance~~
5	Eve	22	HR

02 THE COURSE TABLE

06

```
SELECT department, SUM(credits) AS total_credits
  FROM courses
  GROUP BY department
    HAVING SUM(credits) > 5;
```

Output:

department	total_credits
IT	11

07:

```
SELECT course_id, course_name, department, credits
  FROM courses
  WHERE credits != 4;
```

output course_id course_name department

101	SQL Basics	IT
104	Excel	Finance
105	Statistics	HR

08 SELECT course_id, course_name, credits
FROM students
ORDER BY courses DESC LIMIT 3;

Output: course_id course-name credits

102	Python	4
103	Data Science	4
101	SQL basics	3

03 THE ENROLLMENTS TABLE

09 SELECT MAX(grade) AS Max_grade, MIN(grade) AS min-grade, AVG(grade) AS avg-grade
FROM enrollments

Output

Max_grade	min-grade	avg-grade
90	78	83.4

10 SELECT course_id, COUNT(enrollment_id) AS erollment_count
FROM enrollment
GROUP BY course_id;

Output:

course_id	erollment-count
101	1
102	1
103	1
104	1
105	1

04 THE SALARIES TABLE

11 SELECT department, SUM(salary) AS total-salary, SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;

output

department	total_salary	total-bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12 SELECT department, AVG(salary) AS avg-salary
FROM salaries
GROUP BY department HAVING AVG(salary) > 55000;

Output:

department	avg-salary
IT	61000
finance	70000

13 SELECT employee-id, name, salary, bonus, (salary + bonus) AS total_compensation
FROM salaries WHERE (salary + bonus) > 60000;

output:

employee-id	~~name, salary~~ name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

05 THE PROJECTS TABLE

14 SELECT department, sum(budget) AS total-budget, AVG(budget) AS avg-budget
FROM projects
GROUP BY department HAVING AVG(budget) > 70000

output:

department	total-budget	avg-budget
IT	270000	135000
finance	80000	80000

output Projet.
1
2
5

query: SELEC
FRO
W

output

Projot-id	Project-name	department	budget
1	AI APP	IT	120000
2	payroll System	Finance	80000
5	HR portal	HR	50000

query: SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000 AND department != 'Marketing';